SRI RAMACHANDRA
INSTITUTE OF HIGHER EDUCATION AND RESEARCH
(Category - I Deemed to be University) Porur, Chennai
SRI RAMACHANDRA FACULTY OF ENGINEERING AND TECHNOLOGY

Multi-Agentic AI Assistant using Retrieval-Augmented Generation

AVENZA AI: A Source-Grounded Assistant with a Fine-tuned GPT-2 Answerability Model

Shri Niketh R (E0324071), Koyi Mohith (E0324076), Prof. Deena
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Introduction

Large language models write fluent text but often invent facts, figures and dates. AVENZA AI is a full-stack assistant for finance and technology work that uses the language model only to **write**, while every **fact** comes from retrieved sources: the user's documents, saved memory, a recent-news index and live web search. Answers cite their sources as [1], [2], and twelve specialist agents handle code, math, finance, files, images, diagrams and voice.

Problem Statement

- LLMs hallucinate numbers and facts, which is unsafe for financial and technical decisions.
- Users need answers grounded in their own files (PDF, Word, Excel, slides, scans) and in current data.
- A system must know when the sources do **not** answer a question and say so instead of guessing.

Methodology

1. **Question Understanding** — Fix typos and shorthand, resolve follow-ups, detect intent, entities and recency.
2. **Agent Routing** — Router picks 1 of 12 specialist agents, each with its own instructions and retrieval policy.
3. **Parallel Retrieval + Reranking** — Documents and memory (ChromaDB, nomic-embed-text), RSS news, web search; cross-encoder MiniLM-L12 drops off-topic passages.
4. **Grounded Generation** — LLM writes from numbered sources only. Own GPT-2 (124M) fine-tuned to answer or refuse, with an answerability head.
5. **Verification & Reporting** — Figure check flags numbers absent from sources; classification report page with PDF export.

Architecture and Experimental Setup

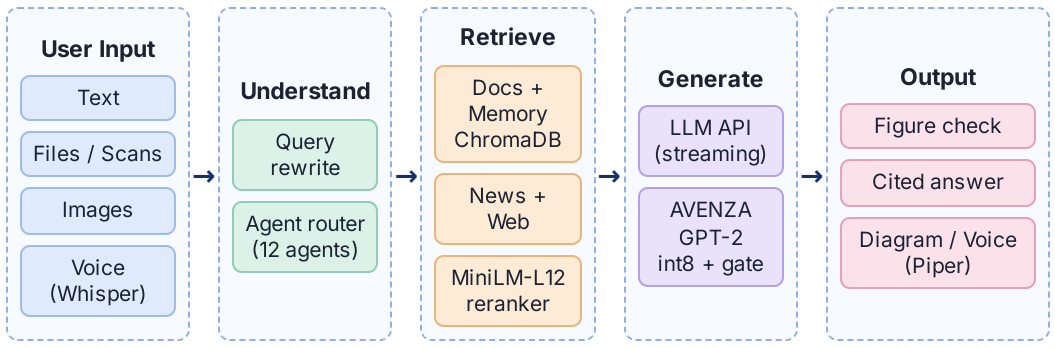


Fig 1. System architecture of AVENZA AI (Next.js web app on :3000, Flask backend on :5000).

Experimental Setup

- **Base model:** GPT-2, 124.4M parameters
- **Datasets:** SQuAD v2, Dolly-15k
- **Split:** 34,013 train · 1,992 val · 300 held-out test
- **Training:** 2 epochs, batch 16, LR 5e-5, max length 640
- **Hardware:** Tesla T4 (Colab), 59.2 min
- **Deployment:** CTranslate2 int8 on CPU
- **Metrics:** Accuracy, Precision, Recall, F1, ROC AUC

Technologies Used

- Python 3.13
- PyTorch + Transformers
- Next.js 16 / React 19
- Flask
- ChromaDB
- Ollama embeddings
- ONNX Runtime
- CTranslate2 int8
- Whisper + Piper
- Google Colab (T4)

Results and Discussion

Grounded Answer with Citations

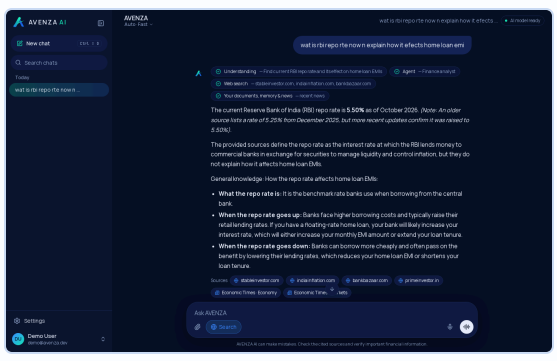


Fig 2. Finance analyst agent: question understood despite typos, web sources cited.

Answerability Classifier

Metric	Colab val (FP32)	Laptop test (int8)
Examples	1,992	300
Accuracy	0.735	0.740
Macro P	0.735	0.738
Macro R	0.738	0.740
Macro F1	0.735	0.739
ROC AUC	**0.815**	**0.831**

Fig 3. int8 quantisation keeps accuracy (threshold 0.68).

Training Loss (Colab, T4)



Fig 4. Val perplexity 1.137 after 2 epochs.

Per-class Scores (Validation)

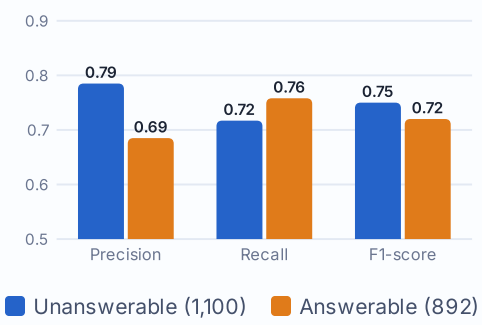


Fig 5. Balanced performance across both classes.

Confusion Matrix (Test, int8)

	Pred: Unans.	Pred: Ans.
True: Unans.	122	43
True: Ans.	35	100

77% answered when answerable · **73%** refused when not · **0.6 s** per answer on CPU

Key Findings

- Fine-tuned GPT-2 detects unanswerable questions with **ROC AUC 0.83** and 74% accuracy on held-out data.
- int8 CTranslate2 runs about **50× faster** than the NumPy engine, with no accuracy loss.
- The full pipeline refuses 73% of unanswerable questions instead of guessing.
- Every answer cites sources; a figure check flags any number not found in them.

Limitations and Future Work

- GPT-2 (124M) gives short answers; long, detailed replies still use an external LLM API.
- About 1 in 4 answerability decisions is wrong; more data or gpt2-medium should help.
- Live web search adds 2–5 s per question; English only for now.
- Future work: larger training set, code model, Wikipedia source and Indian-language translation.

Conclusion

AVENZA AI shows that retrieval, reranking and a fine-tuned answerability model make an assistant that answers from evidence, cites it, and admits when it does not know. Our own GPT-2, trained on SQuAD v2 and Dolly-15k, runs locally in int8 at 0.6 s per answer with ROC AUC 0.83, and the multi-agent design extends the same grounding to files, images, diagrams and voice.

Team Members
Shri Niketh R (E0324071) | Koyi Mohith (E0324076) | Prof. Deena

Guide
Prof. AV Geetha
Sri Ramachandra Faculty of Engineering and Technology, SRIHER

Department
Computer Science and Engineering
Sri Ramachandra Faculty of Engineering and Technology